

一、

1. $a = \frac{M+m}{m} g$

2. $v = \sqrt{\frac{k}{m}} \cdot l_0$ $u = \sqrt{\frac{k}{m_{\min}}} \cdot l_0$

3. 4. 5. 缺题

6.

7. $v_s = 0.25 \text{ m/s}$

8. $\Delta s = \frac{1}{2} g l^2$

9. 10 缺题

11. $\frac{e}{2m}$

12.

二、

1.

2. $m'v l = (\frac{1}{2} m l^2 + m' l^2) \omega$

$$\omega = \frac{m'v}{(\frac{1}{2} m l^2 + m' l^2)} = 15.4 \text{ rad/s}$$

$$-M r = (\frac{1}{2} m l^2 + m' l^2) \beta$$

$$0 = \omega^2 = \beta \theta$$

$$\theta = \frac{(\frac{1}{2} m l^2 + m' l^2) \omega^2}{2 M r} = 15.4 \text{ rad}$$

3. (1) 0处相位较P处提升: $\Delta \varphi = \frac{2\pi}{\lambda} \cdot L$, 即0处振动方程: $y_0 = A \cos(\omega t + \varphi + \frac{2\pi}{\lambda} L)$

(2) 令 $t \rightarrow t + \frac{\lambda}{u}$, 则表达式为: $y(x, t) = A \cos[\omega(t + \frac{\lambda}{u}) + \varphi] = A \cos(\omega t + \frac{\omega \lambda}{u} + \varphi)$

(3) 设P处位置为 x_p , 则 $\omega t + \frac{\omega x_p}{u} + \varphi = \omega t + \frac{\omega \lambda}{u} + \varphi + 2\pi n$ ($n=0, \pm 1, \pm 2, \dots$)

$$\text{即 } x = x_p - \frac{2\pi n \lambda}{\omega}$$

4. 由绝热过程: $V^{r-1} T = C$, 对两部分:

I部分: $V_0^{r-1} T_0 = (\frac{V}{2})^{r-1} T_1$ 则 $T_1 = 2^{\frac{r-1}{r}} T_0$

II部分: $V_0^{r-1} T_0 = (\frac{3V}{2})^{r-1} T_2$ 则 $T_2 = (\frac{2}{3})^{\frac{r-1}{r}} T_0$

$$C_{v,m} = \frac{3}{2} R, \quad C_{p,m} = \frac{3}{2} R + R = (\frac{5}{2} + 1) R, \quad r = \frac{C_{p,m}}{C_{v,m}} = \frac{\frac{5}{2} + 1}{\frac{3}{2}} = 1 + \frac{2}{3}, \quad r-1 = \frac{2}{3}$$

外力做功: $A_1 = \nu C_{v,m} \cdot \Delta T_1 = \frac{3}{2} \nu R T_0 (2^{\frac{2}{3}} - 1)$ (绝热) $A_2 = \nu C_{v,m} \cdot \Delta T_2 = \frac{3}{2} \nu R T_0 [(\frac{2}{3})^{\frac{2}{3}} - 1]$ (绝热)

则外力作的功: $A = A_1 + A_2 = \frac{3}{2} \nu R T_0 [2^{\frac{2}{3}} + (\frac{2}{3})^{\frac{2}{3}} - 2]$

5. $2\pi R L E = \frac{Q}{\epsilon}$ 即 $E = \frac{Q}{2\pi R L \epsilon}$ $\Delta V = \int_0^R E dr = \frac{Q}{2\pi \epsilon L} \ln \frac{R}{r_0}$ 则 $C = \frac{Q}{\Delta V} = \frac{2\pi \epsilon L}{\ln \frac{R}{r_0}}$

若击穿时场强为 $E_0 = \frac{Q}{2\pi R L \epsilon}$, 则此时

$$\Delta U_{\max} = \frac{Q^2}{C} = \frac{2\pi R L \epsilon E_0}{\ln \frac{R}{r_0}} = R E_0 \ln \frac{R}{r_0}$$

6. $d = 500 \text{ nm} = 2 \times 10^{-6} \text{ m}$ $a = b = \frac{1}{2} d = 1 \times 10^{-6} \text{ m}$

(1) $d \sin \theta = k \lambda$ ($k=0, \pm 1, \pm 2, \dots$) $\Rightarrow k = \frac{d \sin \theta}{\lambda}$

$$k_{\max} = \frac{d \sin 90^\circ}{\lambda} = \frac{2 \times 10^{-6}}{5.89 \times 10^{-7}} = 3.4 \quad \therefore k_{\max} = 3$$

$$\therefore \frac{d}{a} = 2 \quad \text{则 } k = \pm 2 \text{ 处缺级, 因此可以观察到 } 0, \pm 1, \pm 3 \text{ 共5条条纹}$$

(2) $d(\sin \theta_0 + \sin \theta) = k \lambda \quad \Rightarrow k_{\max} = \frac{d(\sin 90^\circ + \sin 90^\circ)}{\lambda} = \frac{2d}{\lambda} = 5.09 \quad k_{\max} = 5$

$$\therefore k = \pm 2, \pm 4 \text{ 处缺级, 故可以观察到 } 0, \pm 1, \pm 3, \pm 5 \text{ 共7条条纹.}$$